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Tech Overview

Structural Re-Parameterization, Spatial Coordinate Encoding, and Cross-Domain Robustness for Real-Time Safety Helmet Detection in Construction Surveillance

📍 2027 2nd International Conference on Advances in Artificial Intelligence and Machine Learning (AAIML 2027) - Applied AI and Computer Vision Track

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Summary

This paper proposes Rep-YOLO11s, an edge-friendly safety-helmet detector that integrates three main ideas: structurally re-parameterized neck blocks to reduce runtime memory access cost, a stem CoordConv to inject elevation-aware spatial priors, and Bi-Level Routing Attention (BiFormer) in the neck to suppress background tokens and emphasize tiny objects. The authors pair these with Focal EloU regression and Task-Aligned Assignment to stabilize training under severe class imbalance and show real-time performance gains and solid accuracy on the SHWD benchmark, plus improved zero-shot transfer to Hard Hat Workers, with extensive deployment profiling across GPUs and embedded devices.

Strengths

Technical novelty and innovation

- Integrates structural re-parameterization selectively in the neck (rather than uniformly), a thoughtful hardware-aware design choice targeting MAC/latency bottlenecks.
- Applies CoordConv only at the stem to encode elevation priors with minimal overhead, a simple but domain-relevant trick.
- Uses BiFormer in the neck with practical, plugin-free ONNX/TensorRT deployment, addressing a common barrier for attention on edge devices.
- Decouples objectives via Focal EloU and TAL to alleviate severe class imbalance typical in helmet datasets.

Experimental rigor and validation

- Provides multi-seed ablations, 5-fold cross-validation, and evaluations on both in-domain (SHWD) and out-of-domain (Hard Hat Workers) data.
- Reports detailed latency and energy measurements across multiple platforms (Tesla T4, RTX 3050, Jetson Orin Nano, CPU), with FP32/FP16/INT8 engine variants and clear timing methodology (CUDA events).
- Includes a (limited but useful) INT8 PTQ result with small reported accuracy drop and significant speedup.

Clarity of presentation

- Clear statement of practical deployment constraints and how each module addresses a specific pain point (MAC, tiny objects, background clutter, imbalance).
- Solid mathematical exposition of the re-parameterization process and concise presentation of BiFormer routing mechanics and loss functions.
- Sensible discussion of annotation protocol differences (person box definitions) and their effect on joint metrics vs helmet-only reporting.

Significance of contributions

- Addresses a meaningful applied safety problem with engineering rigor and an eye toward deployability.

- Demonstrates an end-to-end surveillance pipeline and shows that modest architectural changes, chosen with hardware in mind, can deliver large practical benefits.
- The deployment guidance (e.g., ONNX/TensorRT mapping, FPS/J) is of concrete value to practitioners.

Weaknesses

Technical limitations or concerns

- Overall methodological novelty is incremental; the core components (RepConv, CoordConv, BiFormer, Focal EloU, TAL) are known, and the contribution is mainly a careful integration and placement strategy.
- The claimed MAC reduction is inferred from latency; no direct MAC/DRAM bandwidth profiling is presented to substantiate the memory-access claim beyond timing deltas.
- BiFormer's routing can still introduce nontrivial kernel launches/memory movement; more micro-benchmarks (e.g., per-stage operator timings) would help quantify overheads and justify design.

Experimental gaps or methodological issues

- Some reported metrics are confusing or internally inconsistent (e.g., in Table II, A6 "Full Fusion" shows benchmark columns higher than the multi-seed mean by a notable margin), and helmet vs person AP contributions are not comprehensively tabulated per-setting.
- The INT8 PTQ result reports a small mAP50 drop and very large speedup (1.10 ms on T4) but lacks full mAP50-95 retention numbers and per-class effects; the calibration dataset (500 frames) may be small for robust conclusions.
- Cross-paper FPS comparisons rely on literature-reported numbers under different hardware/software stacks; while the paper flags this, the comparisons remain imperfect.
- No release of code/weights/artifacts is indicated, reducing reproducibility of the deployment and quantization results.

Clarity or presentation issues

- The mixture of protocols (joint 2-class vs hat-only, test vs CV, 640 vs 960 resolution) makes it difficult to form a single, apples-to-apples headline number; the narrative could better compartmentalize settings to avoid confusion.
- Some editorial/formatting artifacts (e.g., "VOC2028," table footnotes) and occasional marketing language slightly detract from the scientific tone.

Missing related work or comparisons

- A deeper discussion of NMS-free recent detectors (e.g., YOLOv10 and contemporaries) in the tiny-object occlusion regime would be valuable, as would explicit comparisons with state-space models (e.g., VMamba) on small-object AP.
- Quantization of SR networks can be brittle; relevant recent advances (e.g., outlier-aware BN or QAT for SR nets) could be acknowledged when presenting INT8 results and practical guidance.

Detailed Comments

Technical soundness evaluation

- The structural re-parameterization derivation is sound and standard; confining RepConv to the neck is a reasonable design to avoid early-stage memory blow-up. The paper's claim of reduced MAC is plausible given fewer feature buffer round-trips, and the measured latency improvements on TRT FP16 (−0.50 ms) are consistent with this.
- The CoordConv-in-stem approach is a principled way to inject vertical priors while minimizing runtime cost. Given the domain (helmets above torsos; floor-level distractors), this is well-motivated.
- Incorporating BiFormer in the neck is technically reasonable; maintaining ONNX/TensorRT compatibility via TopK/Gather/Softmax paths is a useful engineering detail. The stated +0.24 ms overhead and improved tiny-object saliency (Grad-CAM illustrations) support the design intent.
- The combined Focal EloU + TAL objective is a standard decoupled solution for skewed class distributions. The argument about CloU's aspect

ratio coupling and vanishing gradients is accurate; EloU's decomposition addresses this.

Experimental evaluation assessment

- The SHWD results are competitive; however, gains over the YOLO11s baseline are modest for joint mAP50-95, and the model appears optimized for helmet recall at some expense to the person class AP (consistent with the paper's explanation about loose boxes and loss penalties). A more granular per-class AP table (50 and 50-95, across 640/960, joint/hat-only) would clarify trade-offs.
- Cross-domain Hard Hat Workers gains without fine-tuning are encouraging, but reliance on helmet-only protocols for the strongest numbers is noted. Reporting both protocols is appreciated; still, explicit per-class APs for both datasets would strengthen claims of robustness.
- The ablation table shows per-component effects and speed deltas; however, the discrepancy between "benchmark" and "multi-seed mean" for A6 deserves clarification. Statistical testing is mentioned (paired t-test), but the hypothesis and tested comparisons are not fully specified.
- Deployment profiling is a real strength: multi-platform timing, FPS/J, and a streaming pipeline breakdown are highly informative for the applied community. The INT8 result is promising but needs fuller accuracy reporting and calibration sensitivity analysis.

Comparison with related work (using the summaries provided)

- Relative to RepVGG/SR literature, this work's novelty is in selective placement and end-to-end deployment focus rather than new re-parameterization theory. That is acceptable for an applied track when supported by careful engineering and evidence.
- Compared to BiFormer (coarse-to-fine routing), the paper reuses BRA in a detector neck context and emphasizes deployment feasibility. It would help to quantify small-object AP deltas attributable specifically to BRA vs CoordConv/RepConv via targeted ablations (APS on COCO-style small bins or SHWD area bins, if available).
- Against recent helmet-focused methods (e.g., DABFNet), the paper offers stronger deployment evidence and cross-domain tests; however, DABFNet reports similar accuracy on SHWD with modest compute. A side-by-side, matched hardware latency comparison (PyTorch and

TensorRT) would be ideal but may be infeasible due to missing code; the paper's caveat about unmatched setups is appropriate.

- The dataset and benchmarking papers (e.g., SFCHD, SH17) highlight domain variability, small-object prevalence, and protocol inconsistencies. The authors' explicit discussion of person-box definitions and their influence on joint scores demonstrates good awareness of these issues.

Discussion of broader impact and significance

- The paper's real contribution is a carefully engineered, deployable pipeline for a societally relevant application (workplace safety), with attention to privacy (on-prem inference) and false-alarm mitigation (helmet-person overlap checks).
- Potential risks (surveillance, bias, labor implications) are partially acknowledged; a deeper treatment (consent, data governance, fairness across skin tones/helmet colors/lighting/time-of-day) would strengthen the ethics discussion.
- The practical deployment notes (RTSP throughput, end-to-end latency budgeting, TensorRT build details) are highly useful for industry practitioners and applied researchers.

Questions for Authors

1. Please clarify the discrepancy in Table II for A6 (Full Fusion): why do the "Benchmark" mAP numbers exceed the "Multi-seed mean" by a substantial margin? Were these measured on different splits or with different evaluation settings?
2. Can you provide per-class AP (mAP50 and mAP50-95) tables for both SHWD and Hard Hat Workers under both joint and hat-only protocols, at 640 and 960 resolutions? This would help verify the trade-off between helmet recall and person AP.
3. For the INT8 PTQ result (1.10 ms on T4), what are the full accuracy metrics (mAP50-95 and per-class AP deltas), and how sensitive are they to calibrator size/content (e.g., 256 vs 500 vs 1000 images)? Any observed failure modes (e.g., small/occluded helmets)?

4. Could you include micro-benchmarks breaking down operator-level timings and memory traffic in the neck (pre/post RepConv fusion and with/without BiFormer) to substantiate the MAC reduction claim beyond end-to-end latency?
5. How robust is the ONNX/TensorRT export for BiFormer across dynamic input sizes and different TensorRT versions/hardware? Any edge cases requiring static shapes or special engine builder flags?
6. Did you consider NMS-free training/inference (as in YOLOv10/TOOD-style dual assignment) in this occlusion-heavy domain, and if so, how did recall behave versus your NMS-based pipeline?
7. Will you release code, ONNX/TensorRT engines, and deployment scripts (including calibration images and engine builder settings) to support reproducibility?

Overall Assessment

This is a solid applied-vision paper focused on a high-impact safety task. While the individual components are known, their selective, hardware-aware integration is thoughtful and well-justified. The authors provide unusually comprehensive deployment evidence across multiple platforms, including energy efficiency and end-to-end streaming throughput, which is valuable for the intended application. Accuracy gains over a strong YOLO11s baseline are modest for joint metrics but meaningful for helmet recall and cross-domain transfer; the paper is transparent about annotation protocol effects and offers a reasonable hat-only evaluation to isolate PPE performance. Some experimental reporting could be clearer (notably the A6 ablation discrepancy, per-class APs, and fuller INT8 accuracy retention), and the work would benefit from publicly released artifacts. Overall, for the AAIML Applied AI and Computer Vision track, the paper offers practical contributions with credible engineering and deployment guidance. I recommend acceptance after addressing the noted clarifications.

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